

Causal Disentanglement of User-Induced and Assigned Persona in Language Models

Ronit Jaliha Bhavya Ahuja Eshaan Sharma Krishak Aneja

1 Motivation

LLMs adopt a persona from assigned system prompt. But it has been observed that this persona drifts over a long conversation. The model starts matching the user’s affect and stylistic traits, known as personality mirroring (Sprejer et al., 2026), and the behavior adopted by system prompt degrades on long, multi-turn interactions (known as persona drift; (Lu, 2026)).

People have seen both. The ways to evaluate these are still naive, like looking at what the model writes, or the whole activation summaries of the model. Thus, till now we do not have a concrete representational mechanism.

Sprejer et al. (2026) leave the underlying mechanism open. They project activations onto trait-space principal components and find statistically significant shifts following user interactions. However, it is unclear what these shifts represent since there are multiple possible interpretations. One possibility is that the model’s internal representation of its own persona changes over the course of the interaction. Another is that the measured shift primarily reflects *personality mirroring*, with the model’s activations reflecting the stylistic and personality cues present in the user’s text as they remain in context. Disentangling those two sources of signal is identified as open work in that paper. We extend this distinction to the question of mechanism: is the shift localized to a specific set of components, or does it arise simply from processing the user’s stylistically shaped context?

Assigned persona is in better shape. Izawa (2026) demonstrated that under residual-stream steering, the persona of the model relies on essentially a very small minority of attention heads, known as Style Modulation Heads (SMHs). Their validation for this was causally using system prompts and steering vectors. But persona that is absorbed from the user has not yet gone through

a similar test. Furthermore, even if the SMH circuit processes both types of personas, a shared computational substrate does not guarantee that the assigned and user-induced signals occupy distinct geometric directions. Along with that, there is no established method to suppress this user-induced style persona drift without degrading the quality of the model’s intended persona.

We therefore ask three questions:

RQ1 Is user-induced persona drift driven by a specific, localized computation, or is it a generalized consequence of processing stylized context without any identifiable locus?

RQ2 Do user-induced mirroring and assigned persona expression share the same SMHs as a unified mechanistic substrate or do they have different causal pathways and heads?

RQ3 Are the assigned and user-induced components of persona geometrically separable within this circuit? And if yes, can we use this separability to suppress user-induced drift while preserving the assistant’s initially designated persona?

2 Related Work

Persona as a Geometric Object. In the recent interpretability work, we found that persona and personality traits are not just superficial artifacts of prompting, but rather occupy structured and continuous directions in the model’s activation space (Chen et al., 2025). However, if we directly intervene on these directions via residual stream steering, we find that it reduces downstream coherence (Subramani et al., 2022). Thus, we need to identify a more localized site for intervention rather than directly modifying the residual stream.

Persona Drift and Mirroring. In literature, we find that models drift from their default assistive

personality over long horizons, and that clamping the activations along the Assistant Axis in its activation space mitigates this drift (Lu, 2026). Separately, we also find that models show user-induced personality mirroring, which can be seen as a drift in the principal components of the trait space (Sprejer et al., 2026), but we could not find papers on whether this drift is in the model’s own persona representation or if it is an artifact of processing styled content. As a preliminary experiment, we compare Assistant Axis activation in user-token processing versus assistant-token generation, offering some evidence that the personality of the model changes before even generating anything (Beckmann and Butlin, 2026). This result is correlational and relies on a coarse axis of the whole model space rather than localising a specific circuit. We propose to test this question causally at the level of an identified set of attention heads.

Circuit-Level Causal Mechanisms. In literature, we find that persona representation is not present across the residual stream but rather highly localised in a certain sparse set of attention heads, called Style Modulation Heads (Izawa, 2026). This helps us identify a causally relevant site where we can look for changes in the persona of the model and intervene to steer it. Previous analysis is only restricted to system-prompt-induced identity and doesn’t look into user-induced signal (Poonia and Jain, 2025). We also find that, though conversational boundaries exist at the token level, they get dissolved in the latent geometry where style features dominate, which explains why linear probes trained across conversational turns confuse genuine persona representations with surface syntax and role position (Ye et al., 2026).

In summary, no work tests whether a single head-level causal circuit is responsible for both assigned and user-induced persona and whether these signals are actually geometrically separable.

3 Dataset

No existing dataset provides multi-turn conversations with independently controlled assigned persona and user-induced style, matched for content and confound-free at the level required for our counterfactual and factorial designs (Sections 4). We therefore construct our own dataset, spanning two variants aligned to the two experimental stages described in Section 4.

Trait and Domain Taxonomy. We adopt the Big Five personality dimensions (Openness, Conscientiousness, Extraversion, Agreeableness, Neuroticism) as our trait taxonomy, rather than model-specific principal components (Sprejer et al., 2026), since Big Five is defined independently of any particular model and transfers directly across Qwen and Llama. For each trait, we define a high-pole and low-pole phrasing style (e.g., for Agreeableness: polite, deferential, hedge-heavy phrasing vs. blunt, demanding phrasing). Base scenarios are drawn from four domains — coding, explaining/understanding, emotional advice, and brainstorming — to ensure effects are not artifacts of a single conversational topic.

Counterfactual Dataset (Stage 1). We write 5 base pairs (user-intent, assistant-response) for each of the four domains, yielding us 20 base scenarios in all. Then, each base scenario is rewritten in the high-pole and low-pole phrasing of all 5 traits, giving us a total of 200 stylistic variants. Then, for each variant we generate 2 additional paraphrases to make sure that signal isn’t a result of a single stale template. In all, this gives us 600 conversations to use.

Each base scenario has a fixed assistant response, and this response is teacher-forced across all stylistic variants, so that the only difference is the user’s phrasing. This provides us with the requisite data for our counterfactual experiments.

Factorial Dataset (Stage 2). For the regression in Section 4, we generate both the **assigned persona** (system prompt instructing high pole, low pole or no persona) and **user style** (high-pole or low-pole per trait). This gives us 6 conditions per trait, which would then be multiplied with 20 base pairs and 5 traits to give us 600 condition - scenario pairs. Since this stage has natural generation, we sample 3 completions per pair to account for sampling variance, giving us 1800 conversations per model and 3600 across both.

Quality Control. To make sure the data is up to standards, we add an LLM judge verification step. An additional judge will evaluate the stylistic variant on two criteria : (i) *content preservation* : does the variant express the same intent as the base scenario, independent of style; and (ii) *trait realization* : does the phrasing clearly express the intended pole for the trait. Variants failing screening are re-generated and are given the feedback generated by

the judge. As a final check, we manually verify a random 10% sample of the final dataset for quality.

4 Proposed Methodology

Our approach has 3 stages: a counterfactual analysis to determine whether user-induced style surfaces in the SMHs, a latent space analysis designed to geometrically plot and disentangle assigned and user-induced persona, and as a final check an intervention to test whether the identified representations play a causal role. To determine whether a property is specific to the SMHs, we compare the SMHs against two control head groups at the same layer. One group is randomly selected and other is selected based on highest activation norm. This will allow us to isolate whether these effects appear at the SMHs specifically or not.

4.1 Counterfactual Setup

We use the constructed counterfactual conversations which have fixed user intent and assistant response, with only the user’s phrasing varying. We need a model independent taxonomy since we are using 2 models, and thus we chose to use the Big Five personality dimensions as our taxonomy, further defining high and low pole phrasing styles for each trait. For example, high and low Agreeableness are expressed through polite, deferential and blunt, demanding phrasing respectively.

For each base scenario, we teacher-force an identical assistant response across all stylistic variants and extract activations from the three head groups at the first assistant token, immediately following the final user token. Thus, changes in activation cannot be attributed to differences in the generated response and instead isolate the effect of the user’s stylistic expression. These controlled measurements form the basis for our analysis of how user style is represented within and outside the SMH circuit.

4.2 Latent-Space Analysis

We next vary both the persona assigned through the system prompt and the user’s expressed style in a factorial design. For each conversation, we collect the activations Z at the same first assistant-token position and model them separately for each head group as

$$Z = \beta_0 + \beta_A X_A + \beta_U X_U + \beta_I (X_A \otimes X_U) + \epsilon,$$

where X_A and X_U are effect-coded representations of the assigned persona and user style across the trait and pole conditions. The coefficients β_A and β_U capture their independent contributions, while β_I captures whether the two sources interact beyond an additive combination.

We use β_A and β_U to construct activation subspaces A and U associated with assigned and user-induced persona, and measure their geometric overlap using principal angles. We compare the variance explained and subspace geometry across SMH and both control groups. The interaction term provides an additional specificity test since differences in its magnitude across head groups can reveal whether the interaction between assigned and user-induced persona is localized to SMH or is a more general property of representations at that layer.

4.3 Targeted Interventions

Finally, we test whether the identified user-induced subspace plays a causal role in personality mirroring. Using the estimated U subspace, we construct a projection that removes activation components along this direction and apply it at SMH during autoregressive generation on held-out drift-inducing conversations. We evaluate whether the intervention reduces user-style mirroring while preserving the assistant’s assigned persona and coherence.

To establish that any effect is specific to SMH rather than a consequence of intervening on arbitrary attention heads, we repeat the identical procedure using the estimated U subspaces from both control groups. Comparing the behavioral effect and coherence cost across all three interventions provides a causal test of whether SMH is a privileged site for selectively suppressing user-induced persona drift. A subset of conversations may additionally be extended across multiple turns, time permitting, to examine whether these effects strengthen as drift accumulates.

5 Evaluation Plan

We evaluate our hypothesis at two levels. The Representational and the Behavioral level. At the representational level, we measure regression explained variance and compute the principal angles between the assigned-persona and user-induced subspaces, and benchmark the SMHs against two layer-matched control head groups. At the behavioral level, we score targeted interventions using GPT-4.1 across three metrics. They are user style

mirroring, persona preservation, and output coherence. The exact same intervention pipeline will be applied across all three head groups. This will ensure that the shifts at arbitrary heads are not mistaken for SMH-specific causal effects. When the user style interacts with the assigned persona heavily, we break down the metrics by condition and report the results across individual persona conditions.

6 Tentative Timeline

September: Finalize dataset collection. Ensure quality control over both the counterfactual and factorial datasets. We also plan on completing early experiments to test the pipeline and pin down the necessary attention heads.

October: Shift to analyzing the latent space. We'll use regression to untangle user-induced drift from the assigned persona, and map out the resulting subspaces and run comparisons against all the control groups we have.

November: Roll out targeted interventions to evaluate how well they maintain the baseline persona and coherence without mirroring the user. If we have time and schedule allows, we'll squeeze in the multi-turn drift analysis before finalizing the paper.

References

- Pierre Beckmann and Patrick Butlin. 2026. Where is the mind? persona vectors and llm individuation. *arXiv preprint arXiv:2604.17031*.
- Runjin Chen, Andy Ardit, Henry Sleight, Owain Evans, and Jack Lindsey. 2025. Persona vectors: Monitoring and controlling character traits in language models. *arXiv preprint arXiv:2507.21509*.
- et al. Izawa. 2026. Steering at the source: Localized style modulation in language models. In *ICLR Re-Align Workshop*.
- et al. Lu. 2026. The assistant axis: Characterizing persona drift in multi-turn dialogue. *arXiv preprint arXiv:2601.10387*.
- Ansh Poonia and Maeghal Jain. 2025. [Dissecting persona-driven reasoning in language models via activation patching](#). In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 24553–24566.
- Eitan Sprejer et al. 2026. White-box monitoring for personality mirroring in conversational ai. In *ICLR Workshop on Representational Alignment*.
- Nishant Subramani, Nivedita Suresh, and Matthew E. Peters. 2022. [Extracting latent steering vectors from pretrained language models](#). In *Findings of the Association for Computational Linguistics: ACL 2022*, pages 566–581.
- Charles Ye, Jasmine Cui, and Dylan Hadfield-Menell. 2026. [Prompt injection as role confusion](#). *Preprint*, arXiv:2603.12277.